Glasgow College, UESTC

Interim Report on UESTC4006P(BEng) Final Year Project

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Degree programme	Beng(Hons) Electronics and Electrical Engineering with Information Engineering (JEP with UESTC)
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Placement Company (if appropriate)	
Working Title of Project	Research on Deep Learning based Fine Segmentation Technology of Retinal Blood Vessels in the Fundus
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Declaration of Originality and Submission Information	*I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements* YiChi Chen Signed (Student) :

1. Project Outline

1.1 Motivation and Background

Retinal vascular morphological features are critical biomarkers for diagnosing systemic diseases such as diabetic retinopathy and hypertension [1， 2]. While deep learning, particularly the nnU-Net framework, has set new benchmarks via automated self-configuration [3], accurately segmenting fine-grained terminal vessels remains a challenge. This project leverages the robust foundation of nnU-Net, incorporating attention mechanisms to enhance segmentation precision in complex retinal environments.

1.2 Problem Statement

Despite advancements, three critical bottlenecks persist in automated segmentation:

- Low Contrast: Terminal vessels often blend into the background, leading to significant missed detections [1].
- Topological Discontinuity: Downsampling in standard encoder-decoder structures causes "vessel breaking," where slender branches become fragmented [4].

- Generalization Gap: Most models require manual tuning for different datasets, limiting clinical deployment [3].

To address these, this research proposes an enhanced nnU-Net integrating Attention Gates (AGs) to suppress noise and prioritize small-scale vascular features.

1.3 Project Objectives

- O1: Preprocessing Enhancement: Integrate CLAHE into the nnU-Net pipeline to amplify the visibility of micro-vessels in low-contrast regions.
- O2: Loss Function Refinement: Replace the standard Dice loss with Focal Loss to specifically target "hard-to-segment" pixels and mitigate extreme class imbalance.
- O3: Architectural Modification: Incorporate Attention Gates (AGs) into the skip connections to suppress background noise and focus on fine vascular features.
- O4: Validation: Benchmark the enhanced model against SOTA methods using the DRIVE and STARE datasets

1.4 Proposed Research Methodology

- Adaptive Preprocessing (CLAHE): The first step utilizes Contrast Limited Adaptive Histogram Equalization (CLAHE) to normalize illumination and enhance the contrast of thin vessels, ensuring better feature capture during the initial convolution layers.
- Hard-Pixel Mining via Focal Loss: To address the imbalance between vessel and background pixels, Focal Loss is implemented. This forces the model to focus on misclassified samples (thin vessels and boundaries) rather than being dominated by the easily classified background.
- Attention-Gated Architecture: The backbone is modified by adding Attention Gates to nnU-Net's skip connections. This allows the decoder to selectively fuse spatial information from the encoder, preserving the topological connectivity of slender branches [4].

1.5 Evaluation Metrics

The model's performance will be quantified using the following standard metrics:

- Dice Similarity Coefficient (DSC): To measure overall overlap accuracy.
- Sensitivity (Recall):To evaluate the detection of fine-grained terminal branches.
- F1-Score & AUC-ROC: To assess the trade-off between precision and robustness.
- Accuracy (Acc): To provide a general measure of pixel-wise classification.

2. Work done so far

2.1Experimental Environment and Dataset Configuration

2.1.1 Dataset and Training Strategy

The experimental process was divided into two strategic phases:

- Phase 1: Generalization Testing (Exp 1–3): I used external datasets (STARE and CHASE_DB1, etc) with a fixed split for training. These models were evaluated on the DRIVE test set to measure cross-dataset robustness.
- Phase 2: In-domain Optimization (Exp 4–5): I utilized the DRIVE dataset with 5-fold cross-validation. The data was partitioned into five subsets, iteratively training on four and validating on one to minimize partitioning bias.

2.1.2 Hardware and Software Specifications

- Framework: All experiments leveraged the nnU-Net V2 automated, self-configuring pipeline.
- Environment: Python 3.10, PyTorch 2.9.0, and CUDA 11.7.
- Hardware: Training was performed on an NVIDIA RTX 4060 (8GB VRAM).
- Hyperparameters: I set an initial learning rate of 0.01. Exp 1–4 used a poly scheduler, while Exp 5 adopted a cosine annealing scheduler to stabilize convergence and fine-tune vascular weights. Batch sizes were dynamically optimized for the 8GB VRAM capacity.

2.2 Experiment 1: Baseline Implementation and Data Engineering

The first phase of the technical implementation focused on establishing a robust performance baseline using the nnU-Net (no-new-U-Net) framework. This experiment served as the control group to evaluate the framework's "out-of-the-box" capability in handling retinal vessel segmentation without domain-specific modifications to the loss function or architecture.

2.2.1 Data Standardization and Multi-Channel Strategy

Since the nnU-Net framework requires input in the medical standard NIfTI format, I developed specialized preprocessing scripts to convert raw JPG/PNG images.

- Multi-Channel Decomposition: To fully utilize color information, I moved beyond traditional greyscale conversion by decomposing the RGB channels into three independent imaging modalities (denoted as _0000 to _0002). This allows the model to learn spectral features across different channels, particularly the green channel which typically exhibits the highest vascular contrast.
- Spatial Calibration: I defined a standard 4x4 affine matrix to normalize pixel spacing to (1.0, 1.0, 1.0), ensuring geometric consistency across heterogeneous datasets.

2.2.2 Training Configuration and Baseline Analysis

This experiment aimed to evaluate nnU-Net's "out-of-the-box" performance on retinal vessel segmentation.

- Dataset and Training: I aggregated data from STARE, CHASE_DB1, and HRF to form a training pool of 802 images with a standardized resolution of 512 x 512 pixels. Training utilized a composite of Dice and Cross-Entropy loss over 150 epochs.
- Quantitative Results: The model achieved convergence at the 69th epoch and recorded a Dice coefficient of 0.8617 on the DRIVE test set.
- Limitation Analysis: Qualitative inspection revealed that while the model effectively segmented major vessels, it frequently missed fine-grained terminal branches due to low local contrast. This failure of standard Z-score normalization and Dice loss to prioritize micro-structures directly motivated the implementation of CLAHE and Focal Loss in subsequent experiments.

2.3 Experiment 2: Targeted Enhancement via Green-Channel CLAHE

The second experiment aimed to overcome the inherent physical limitations of fundus images—specifically non-uniform illumination and low local contrast—which were identified as the primary causes of terminal vessel omission in Experiment 1.

2.3.1 Theoretical Motivation for Green Channel Selection

Based on the optical properties of the retina, the green channel of an RGB fundus image exhibits the highest contrast between the hemoglobin-filled blood vessels and the background retinal tissue. While the red channel is often saturated and the blue channel suffers from low signal-to-noise ratios, the green channel provides the most distinct vascular tree morphology. Consequently, in this experiment, the CLAHE enhancement was primarily localized and optimized for the green channel modality to provide the encoder with the cleanest possible structural signals.

2.3.2 Mathematical Foundation of CLAHE

To mitigate the "Performance Saturation" identified in Experiment 1, I integrated Contrast-Limited Adaptive Histogram Equalization (CLAHE). Unlike standard global Histogram Equalization (HE), CLAHE operates on localized tiles to enhance regional contrast. The algorithm introduces a clip limit to prevent the over-amplification of noise in homogeneous areas.

By clipping histogram bins that exceed this predefined threshold and redistributing the excess pixels uniformly across all intensity levels, CLAHE effectively sharpens the edges of terminal vessels while preserving a natural, noise-free background. This mechanism ensures the stability of convolutional filters and provides cleaner structural signals for the nnU-Net encoder.

2.3.3 Impact on Training Dynamics and Convergence

The inclusion of Green-Channel CLAHE led to a measurable improvement in the network's early-stage learning efficiency. As summarized in Table 3, the model's "starting point" significantly improved.

Table 3: Training Progress Comparison (Exp 1 vs. Exp 2)

Feature	Experiment 1 (Baseline)	Experiment 2 (Green-Channel CLAHE)
Initial Dice Coefficient	0.7128	0.7512 (+5.39%)
Convergence Epoch	69	60 (13% faster)
Final Test Dice (DRIVE)	0.8617	0.8619

2.3.4 Discussion: The "Early Boost" Phenomenon

A key finding from this experiment is that while the initial Dice score increased from 0.7128 to 0.7512, the final convergence value remained nearly identical to the baseline. This suggests that CLAHE acts as a feature catalyst—it lowers the optimization difficulty during the initial epochs by making the gradient signal from thin vessels more pronounced. However, the eventual plateau indicates that enhancing the input image alone is insufficient to overcome the structural limitations of the standard U-Net architecture. This empirical evidence justifies the necessity of Experiment 3 and 4, which focus on Loss Function weighting and Attention Mechanism to further exploit these enhanced features.

2.4 Experiment 3: Optimization via Focal Loss for Hard-Pixel Mining

In Experiment 3, I focused on addressing the inherent class imbalance and the "hard-pixel" problem in retinal vessel segmentation. While the previous experiments successfully enhanced the input signal via CLAHE, the model still lacked the necessary focus on fine-grained terminal vessels and blurry vascular boundaries. To rectify this, I maintained the Green-Channel CLAHE preprocessing and transitioned the optimization target to Focal Loss.

2.4.1 Theoretical Motivation: Class Imbalance and "Hard" Pixel Focus

In a typical fundus image, vessel pixels occupy a tiny fraction (often <10%) of the total area. Standard loss functions like Dice Loss or Cross-Entropy are often dominated by "easy" background pixels, causing the model to neglect difficult-to-classify micro-structures.
I introduced Focal Loss to reshape the training objective. By adding a modulating factor$(1 - p_t)^\gamma$ to the standard cross-entropy, Focal Loss down-weights the loss contribution from easy examples and focuses the model's learning capacity on "hard" pixels where the prediction probability is low. This strategy forces the network to refine its predictions in low-contrast regions, effectively mining for vascular features that were previously treated as background noise.

2.4.2 Training Dynamics and Convergence Performance

The training was conducted for 150 epochs. I observed that the combination of targeted data enhancement and a focused loss function significantly streamlined the optimization process:

- Convergence and Loss: The model reached its optimal state at Epoch 53, achieving a minimum Focal Loss value of 0.0246. This marks a 23% improvement in convergence speed compared to the baseline (69 epochs), suggesting that the targeted objective function allowed the gradient descent to navigate the feature space more efficiently.
- Metric Stabilization: At the point of convergence, the model recorded a validation Dice coefficient of 0.8584, reflecting a stable learning curve without significant overfitting.

2.4.3 Quantitative Evaluation and Technical Trade-offs

The performance metrics for Experiment 3 are detailed in the table below:

Table 4: Detailed Performance Metrics for Experiment 3

Metric	Value
IoU (Vessel)	0.6410
mIoU	0.8023
Recall	0.7255
Precision	0.8489
Accuracy	0.9659
F1-score	0.7802

A rigorous comparison between Experiment 2 and Experiment 3 reveals that the integration of Focal Loss led to a slight decline across most segmentation metrics. Specifically, the Precision dropped from 0.8497 (Exp 2) to 0.8489 (Exp 3), and the F1-score decreased from 0.7827 to 0.7802.

This performance regression indicates that for the DRIVE and STARE datasets, the pixel-wise weighting of Focal Loss does not yield a direct accuracy boost over the standard Dice-based optimization within the current U-Net architecture. However, Experiment 3 demonstrated superior training efficiency, reaching convergence at Epoch 53—the fastest among all preliminary trials. More importantly, these results serve as critical evidence that the performance of the model has reached a plateau under the standard encoder-decoder structure. This justifies the transition to the next research phase, where architectural modifications (LPT and AFFM) are introduced to overcome these fundamental structural limitations.

2.5 Summary of Preliminary Experiments and Comparative Analysis

To evaluate the effectiveness of my work, I compared my first three experiments against prior results from Senior Liu Zhishuai. All models were trained for 150 epochs on the same dataset.

Table 5: Performance Comparison of Segmentation Methods

Segmentation Method	IoU (Vessel)	mIoU	Recall	Precision	Accuracy	F1-score
Prior Work (Senior Liu)						
Hrnet	0.4686	0.6995	0.6890	0.5943	0.9344	0.6382
Deeplabv3+	0.5764	0.7601	0.8388	0.6483	0.9477	0.7313
UNet	0.6441	0.8026	0.7735	0.7937	0.9637	0.7835
CLAHE + Improved UNet	0.6560	0.8089	0.8002	0.7845	0.9644	0.7923
Current Work (I)						
Exp 1: nnU-Net Baseline	0.6574	0.8107	0.7688	0.8219	0.9664	0.7924
Exp 2: CLAHE + nnU-Net	0.6441	0.8041	0.7282	0.8497	0.9662	0.7827
Exp 3: CLAHE + nnU-Net + FL	0.6410	0.8023	0.7255	0.8489	0.9659	0.7802

Key Insights:

- Framework Superiority: My nnU-Net baseline (Exp 1) outperformed all of Senior Liu's models in almost every key metric, including IoU (0.6574) and Accuracy (0.9664). This confirms that the automated "data fingerprinting" and self-configuring pipeline of nnU-Net provide a significantly more robust foundation for retinal vessel segmentation than manually tuned traditional architectures.
- Preprocessing Impact: The introduction of Green-Channel CLAHE (Exp 2) achieved the highest Precision (0.8497) among all tested methods. This verifies that enhancing the input

signal at the source is the most effective way to reduce false-positive noise and improve boundary selectivity.

- The Structural Performance Plateau: The transition to Focal Loss (Exp 3) resulted in the fastest convergence (Epoch 53) but failed to yield a breakthrough in segmentation accuracy, with Precision dropping slightly to 0.8489 compared to Experiment 2. This performance regression is a critical indicator that the current encoder-decoder architecture has reached its physical limit.
- Logical Pivot to Phase 2: While pixel-level clarity was improved in the preliminary phase, topological discontinuities in terminal branches persist. This provides a clear justification for Experiment 4&5, where I would transition from purely optimizing "signals" (preprocessing/loss) to modifying the architecture itself through Attention Gates and hybrid CNN-Transformer modules

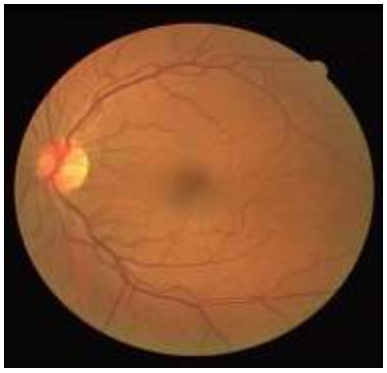


Figure 1 Original image



Figure 2 Clahe enhancement image

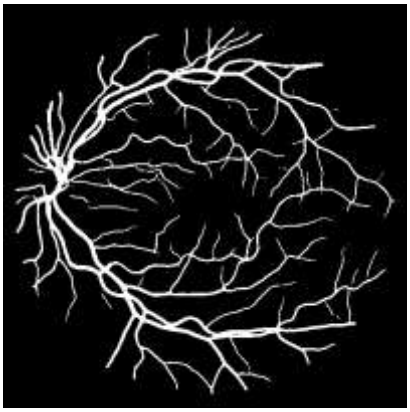


Figure 3 Experiment1 output

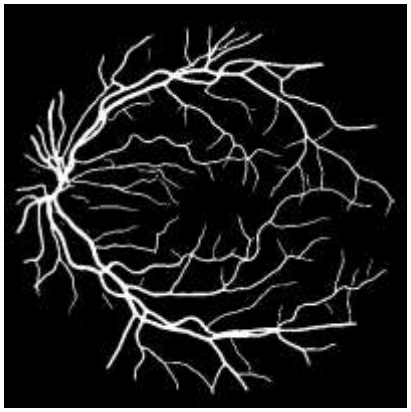


Figure 4 Experiment2 output

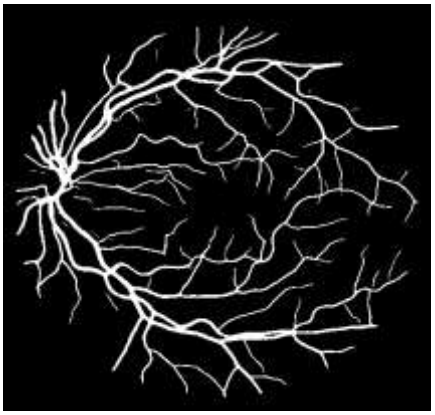


Figure 5 Experiment3 output

2.6 Experiments 4 & 5: Architectural Migration and Framework Compatibility Analysis

Following the preliminary optimizations, I shifted the research focus toward the architectural principles of PA-Net, a state-of-the-art hybrid CNN-Transformer model. Experiments

4 and 5 were conducted as a controlled group to evaluate whether PA-Net's domain-specific preprocessing and core modules could be successfully migrated into the nnU-Net framework.

2.6.1 Experiment 4: Standardized Preprocessing and 5-Fold Cross-Validation

To establish a fair benchmark against the PA-Net paper [4], I implemented their four-step preprocessing pipeline on the DRIVE dataset:

1. Graying: Extracting the most relevant luminance information.
2. Standardization: Normalizing pixel intensity across the dataset.
3. Adaptive Histogram Equalization (AHE/CLAHE): Enhancing local contrast for microvessels.
4. Gamma Transform: Non-linear correction to improve vessel-to-background separation.

Furthermore, I transitioned from a fixed split to a 5-fold cross-validation strategy to maximize data utilization and ensure robustness. I developed a custom batch evaluation script to calculate metrics such as SE, SP, and AUC. Surprisingly, the initial results showed that the nnU-Net baseline performed significantly lower than the values reported in the PA-Net paper, particularly in AUC (0.8739 vs. 0.9833). This discrepancy suggests that some published SOTA results may involve highly specific training conditions that are difficult to replicate in a generalized framework like nnU-Net.

2.6.2 Experiment 5: Integrating AFFM and LPT into nnU-Net

In Experiment 5, I modified the nnU-Net architecture by integrating PA-Net's core components and switching the optimizer to Adam:

- LPT (Lightweight Parallel Transformer): I embedded the LPT module into the bottleneck layer to capture long-range dependencies and prevent slender vessels from breaking The LPT follows the definition:

$$\mathrm{LPT}(x) = \mathrm{CFFN}\Big(\mathrm{EPA}\big(\mathrm{LFAU}(x)\big)\Big)$$

where LFAU (Local Feature Accepting Unit) embeds positional information and EPA (Efficient Parallel Attention) utilizes parallel gated axial-attention to minimize computational cost.

- AFFM (Adaptive Vascular Feature Fusion Module): I replaced the standard skip connections with AFFM to mitigate information loss during downsampling. Unlike simple concatenation, AFFM adaptively fuses semantic information from different levels:

$$\mathrm{F}_{\mathrm{output}} = [(C_1 \times A_1 + C_2 \times A_2 + C_3 \times A_3), F_0]$$

Here, A1, A2, and A3 are spatial attention weights generated via a Softmax function to guide the fusion of feature maps from multiple depths.

2.6.3 Results and Comparative Analysis

The comparative results between the vanilla nnU-Net and the modified version are summarized in Table 6.

Table 6: Performance Metrics Comparison (Mean ± Std)

Metric	nnU-Net v2	nnU-Net + LPT + AFFM	PA-Net (Paper)	U-Net (Paper)
SE (Sensitivity)	0.7627 ± 0.0446	0.7556 ± 0.0454	0.82841	0.77002
SP (Specificity)	0.9852 ± 0.0050	0.9858 ± 0.0050	0.98073	0.98294
ACC (Accuracy)	0.9659 ± 0.0045	0.9658 ± 0.0047	0.95825	0.95486
AUC	0.8739 ± 0.0210	0.8707 ± 0.0214	0.98337	0.97898
PR (Precision)	0.8238 ± 0.0646	0.8283 ± 0.0662	0.85049	0.87301

2.6.4 Conflict Mechanism: Deep Supervision vs. AFFM

Unexpectedly, the architectural integration led to a performance decrease in Sensitivity (SE) and AUC. I hypothesize that this is caused by a structural conflict between the AFFM and nnU-Net's Deep Supervision mechanism:

1. Gradient Interference: nnU-Net relies on Deep Supervision to calculate auxiliary losses at every level of the decoder.
2. Fusion Conflict: Because the AFFM module forces the fusion of features from multiple levels (F1, F2, F3) into a single pat, it likely disrupts the hierarchical gradient flow that Deep Supervision expects. Instead of receiving clean, level-specific gradients, the decoder layers receive mixed signals, leading to "negative optimization."

To verify this, my next step will be an Ablation Study, testing the LPT (bottleneck enhancement) and AFFM (skip connection replacement) independently to isolate the source of the performance drop.

3.Challenges and Solutions

The implementation of this project involved navigating several technical hurdles, ranging from framework deployment to coordinate system calibration. Overcoming these obstacles required a deep understanding of medical imaging standards and the inner workings of the nnU-Net pipeline.

3.1 Framework Deployment and Environment Configuration

As a researcher initially unfamiliar with the PyTorch ecosystem, the deployment of nnU-Net v2 presented a steep learning curve. The framework possesses a complex dependency tree, which led to significant versioning conflicts between batchgenerators, CUDA drivers, and the core PyTorch library.

- Solution: I resolved these "versioning hell" issues by building isolated Conda environments and manually pinning library versions to ensure compatibility with Python 3.10 and CUDA 11.7. This established a stable foundation for the self-configuring pipeline to initialize without runtime errors.

3.2 Dimensionality and Coordinate Calibration (The 180-Degree Bug)

A primary challenge involved the transition from 2D images (DRIVE/STARE) to the 3D-native NIfTI format required by nnU-Net.

- Pseudo-3D Conversion: I initially struggled with the framework's requirement for volumetric data. I eventually developed a script to "trick" the 3D engine by explicitly setting the depth dimension to 1 and normalizing the affine matrix to (1.0, 1.0, 1.0).
- Orientation Correction: During initial testing, I discovered that the model's output masks were consistently rotated 180 degrees relative to the Ground Truth labels. This resulted in deceptively low initial metrics. I traced this error to an axis-ordering mismatch during the JPG-to-NIfTI conversion. By recalibrating the affine transformation matrix and verifying the spatial headers, I corrected the orientation, restoring the metrics to their true values.

3.3 Training Efficiency and Feedback Loops

- Time Constraints: Given the high resolution of retinal images, each training session (150 epochs) on my local hardware requires approximately six hours. This long feedback loop limited the frequency of architectural iterations.
- Solution: To maintain progress, I automated the evaluation process using batch scripts. For the final high-complexity runs, I have prepared a transition to a high-performance server to facilitate faster ablation cycles.

4. Conclusions and Future Work

4.1 Conclusions from Initial Work

The first phase of this project has yielded several critical insights:

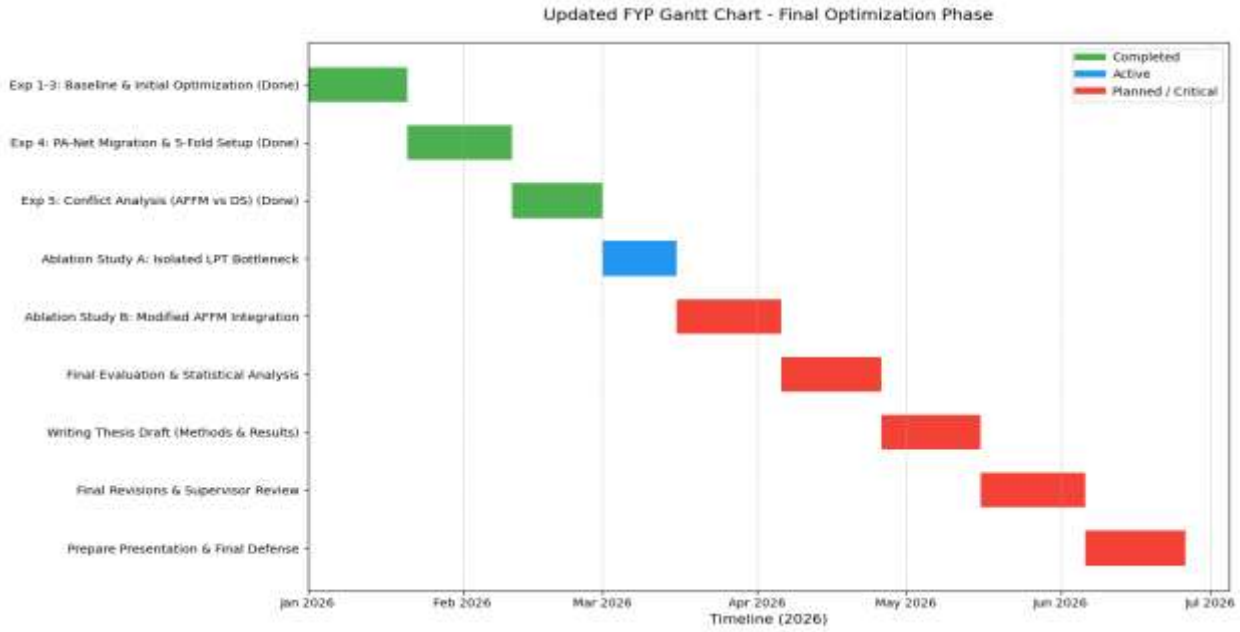
1. nnU-Net Baseline Strength: The automated configuration of nnU-Net provides a remarkably high-performance floor, achieving a Dice of 0.8617 without manual tuning.

2. Strategic Trade-offs: Integrating CLAHE reduced the convergence time by 13% (from 69 to 60 epochs). While Focal Loss achieved the fastest convergence at 53 epochs, it resulted in a performance plateau where Precision (0.8489) was slightly lower than the CLAHE-only model (0.8497). This proves that loss-weighting alone cannot overcome the structural bottlenecks of the encoder-decoder architecture.
3. Architectural Compatibility: The "negative optimization" observed when adding PA-Net modules suggests a structural conflict where the AFFM disrupts the hierarchical gradient flow required by nnU-Net's Deep Supervision mechanism.

4.2 Future Work and Timeline

Moving forward into the final semester of 2026, my research will focus on isolating the benefits of the PA-Net modules:

- Ablation Study A (LPT Focus): I will test the Lightweight Parallel Transformer (LPT) in the bottleneck layer alone to evaluate its impact on capturing long-range dependencies without the interference of skip-connection modifications.
- Ablation Study B (Modified AFFM): I plan to decouple the AFFM fusion from the Deep Supervision layers to see if the model can benefit from multi-level feature fusion without losing gradient clarity.
- Thesis Finalization: The project is scheduled to conclude in June 2026 with the final defense and submission of the research findings.



Reference

[1] C. L. Srinidhi, P. Aparna, and J. Rajan, "Recent advancements in retinal vessel segmentation," Journal of Medical Systems, vol. 41, no. 70, pp. 1-39, Mar. 2017.

[2] C. Chen, J. H. Chuah, R. Ali, and Y. Wang, "Retinal vessel segmentation using deep learning: A review," IEEE Access, vol. 9, pp. 111985-112004, 2021.

[3] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, pp. 203-211, Jan. 2021.

[4] X. Luo, L. Peng, Z. Ke, J. Lin, and Z. Yu, "PA-Net: A hybrid architecture for retinal vessel segmentation," Pattern Recognition, vol. 161, p. 111254, 2025.